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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL JUPYTER
TERMINAL powershell - day6
PS G:\vs\python> python preprocessing.py
PS G:\vs\python> cd day6
PS G:\vs\python\day6> python preprocessing.py
(55000, 13)
Index(['CID', 'TID', 'Gender', 'Age Group', 'Purchase Date',
       'Product Category', 'Discount Availd', 'Discount Name',
       'Discount Amount (INR)', 'Gross Amount', 'Net Amount',
       'Purchase Method', 'Location'],
      dtype='object')
CID          int64
TID          int64
Gender       object
Age Group    object
Purchase Date object
Product Category object
Discount Availd object
Discount Name object
Discount Amount (INR) float64
Gross Amount float64
Net Amount   float64
Purchase Method object
Location     object
dtype: object
<class 'pandas.core.frame.DataFrame'>
```

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL JUPYTER
TERMINAL powershell - day6
PS G:\vs\python\day6> python preprocessing.py
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55000 entries, 0 to 54999
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   CID                    55000 non-null  int64
1   TID                    55000 non-null  int64
2   Gender                 55000 non-null  object
3   Age Group              55000 non-null  object
4   Purchase Date          55000 non-null  object
5   Product Category       55000 non-null  object
6   Discount Availd        55000 non-null  object
7   Discount Name          27415 non-null  object
8   Discount Amount (INR)  55000 non-null  float64
9   Gross Amount           55000 non-null  float64
10  Net Amount             55000 non-null  float64
11  Purchase Method        55000 non-null  object
12  Location               55000 non-null  object
dtypes: float64(3), int64(2), object(8)
memory usage: 5.5+ MB
None
```

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL JUPYTER
TERMINAL powershell - day6
PS G:\vs\python\day6> python preprocessing.py
CID          TID          Gender    Age Group    Purchase Date    ... Discount Amount (INR) Gross Amount    Net Amount    Purchase Method    Location
0   943146  5876328741  Female    25-45        30/08/2023 20:27:08    ...          64.38      725.304000    661.004000    Credit Card    Ahmedabad
1  180079  1018503182   Male      25-45        23/02/2024 00:33:46    ...         175.19    4630.901075    4463.001075    Credit Card    Bangalore
2  337500  3014002210   Other     60 and above  06/03/2022 09:09:50    ...         211.54    1906.372575    1774.832575    Credit Card    Delhi
3  180333  1305204173   Other     60 and above  04/11/2020 04:41:57    ...           0.00    5695.612650    5695.612650    Debit Card    Delhi
4  447553  8009390577   Male      18-25        31/05/2022 17:00:32    ...          439.92    2292.651500    1852.731500    Credit Card    Delhi

[5 rows x 13 columns]
datetime64[ns]
CID          0
TID          0
Gender       0
Age Group    0
Purchase Date 0
Product Category 0
Discount Availd 0
Discount Name 27505
Discount Amount (INR) 0
Gross Amount 0
Net Amount    0
Purchase Method 0
Location      0
Year          0
Month         0
```

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TERMINAL
PS G:\vs\python\day6> python preprocessing.py
Day
dtype: int64
CID
0.000000
TID
0.000000
Gender
0.000000
Age Group
0.000000
Purchase Date
0.000000
Product Category
0.000000
Discount Availed
0.000000
Discount Name
50.154545
Discount Amount (INR)
0.000000
Gross Amount
0.000000
Net Amount
0.000000
Purchase Method
0.000000
Location
0.000000
Year
0.000000
Month
0.000000
Day
0.000000
dtype: float64
0
```

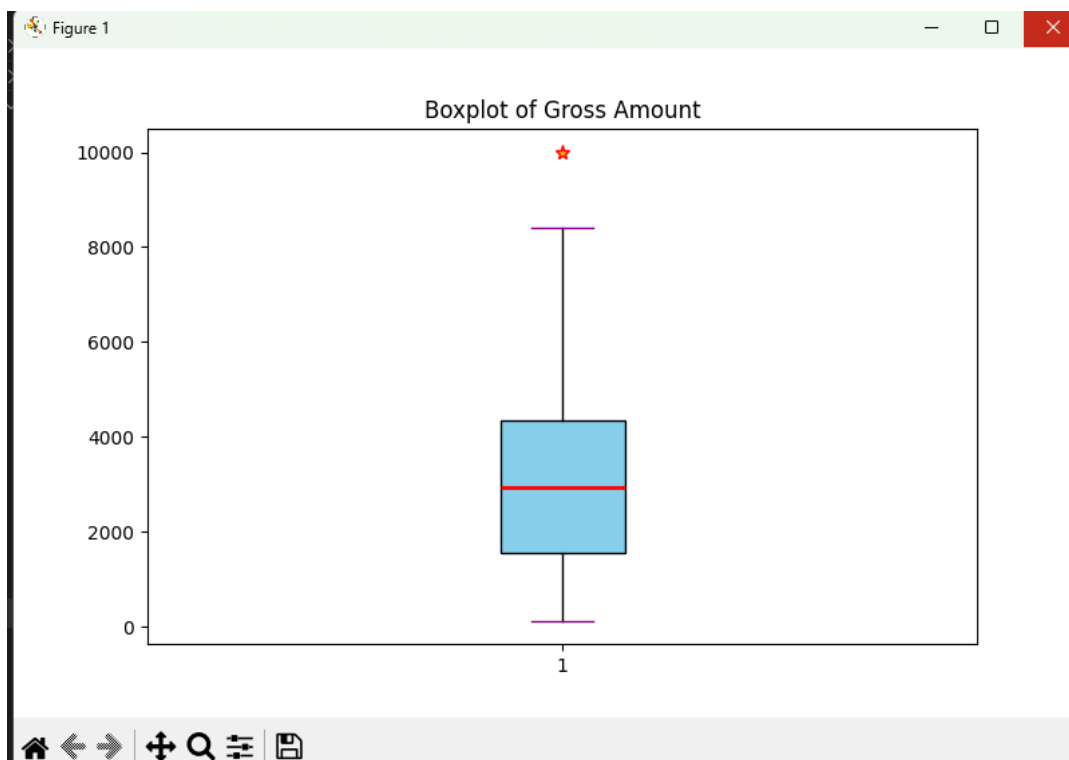
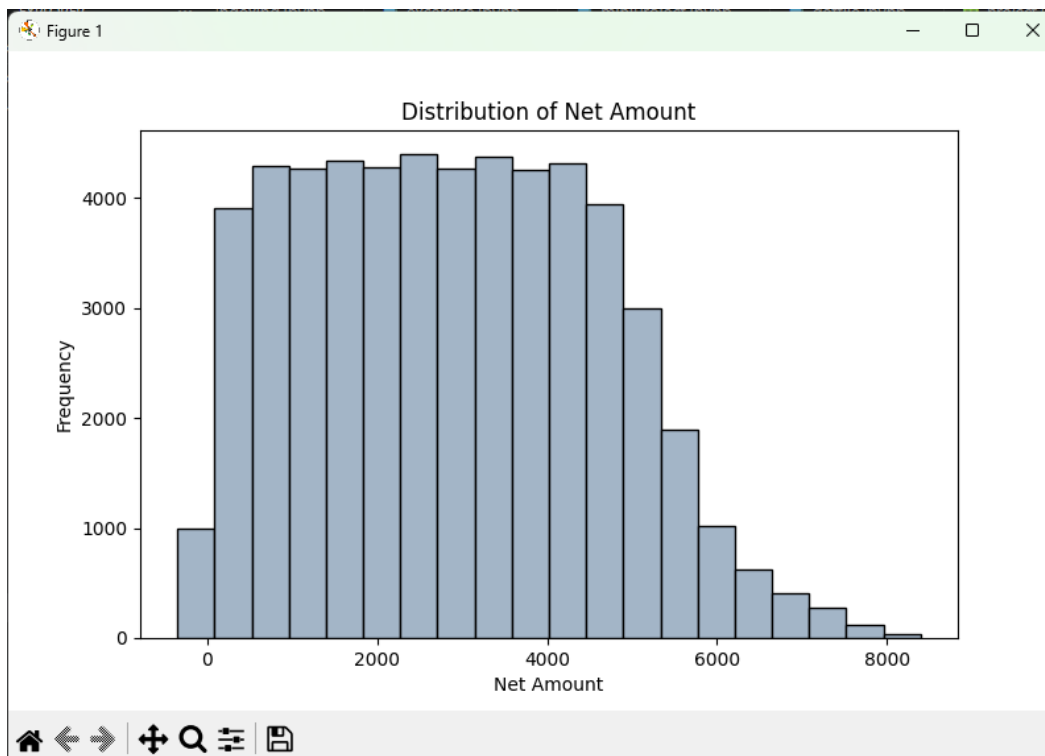
```
TERMINAL
PS G:\vs\python\day6> python preprocessing.py
0
CID      TID      Purchase Date  Discount Amount (INR)  ...  Net Amount  Year  Month  Day
count  55000.000000  5.500000e+04  55000  55000.000000  ...  55000.000000  55000.000000  55000.000000  55000.000000
mean   551245.593891  5.504740e+09  2022-03-19 14:27:22.813054464  136.986796  ...  2875.949810  2021.713527  6.512764  15.766189
min    100009.000000  1.000163e+09  2019-09-19 11:46:07  0.000000  ...  -351.119775  2019.000000  1.000000  1.000000
25%    323717.000000  3.252604e+09  2020-12-17 14:39:58  0.000000  ...  1429.551863  2020.000000  4.000000  8.000000
50%    550088.500000  5.490383e+09  2022-03-17 06:27:07  0.000000  ...  2814.910875  2022.000000  7.000000  16.000000
75%    776955.750000  7.747933e+09  2023-06-21 10:14:07.500000  274.115000  ...  4211.407838  2023.000000  10.000000  23.000000
max    999996.000000  9.999393e+09  2024-09-18 16:00:08  500.000000  ...  8394.825600  2024.000000  12.000000  31.000000
std    208061.330337  2.594534e+09  NaN  165.375502  ...  1726.127778  1.484272  3.447151  8.820393

[8 rows x 9 columns]
Mean Gross Amount = 3012.036685973182
Mean Net Amount = 2875.949810155
Mean Discount Amount = 136.98679681818183
Median of Gross Amount 2954.2661500000004
Median of Net Amount 2814.910875
Std of Gross Amount 1710.4318656418672
Std of Net Amount 1726.1277780390761
Minimum of Gross Amount 136.4543240000000000
Maximum of Gross amount 8394.8256
```

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TERMINAL
PS G:\vs\python\day6> python preprocessing.py
Gross Amount  Discount Amount (INR)  Net Amount
1.000000      0.001473             0.998527
Discount Amount (INR)  0.001473             1.000000  -0.001473
Net Amount  0.998527             -0.001473  1.000000
Gender
Female  18454
Other  18450
Male  18096
Name: count, dtype: int64

Purchase Method
Credit Card  22096
Debit Card  13009
Net Banking  5485
International Card  2815
Cash on Delivery  2768
PhonePe UPI  2683
Paytm UPI  2674
Google Pay UPI  2670
Name: count, dtype: int64
```

```
TERMINAL
PS G:\vs\python\day6> python preprocessing.py
Location
Mumbai  11197
Delhi  10799
Bangalore  8249
Hyderabad  5545
Chennai  4368
Pune  3781
Ahmedabad  2785
Kolkata  2789
Jaipur  1678
Lucknow  1136
Other  1046
Varanasi  606
Dehradun  553
Srinagar  548
Name: count, dtype: int64
```



```
Lower limit = -2607.9815624999999
Upper limit = 8512.5991375
Q1 = 1562.2362
Median = 2954.4383500000004
Q3 = 4342.381375
```

